Attention models are loosely inspired by human and animal forms of attention. With human vision, we cannot see or focus on the entire field of view in front of us; our eyes make saccadic (rapid, jerky) movements to scan across the field of view, and we can consciously decide to focus on a particularly salient area within our view. This allows us to focus on processing the relevant aspects of a scene, which is an efficient use of resources. Moreover, when we're engaged in thought and reasoning, we can only attend to a few things at once. We also naturally tend to employ relational reasoning when we say things like "he is older than her" or "the door closed behind me;" we are relating the properties or behavior of certain objects in the world to others. Indeed, words in human language generally only convey meaning when related to other words. In many cases, there is no absolute frame of reference; we can only describe things as they relate to other things that we know.

Absolute (nonrelational) attention models are designed to function like our eyes in that they try to learn how to extract only the relevant parts of the input data for efficiency and interpretability (you can see what the model is learning to attend to when making a decision), whereas the self-attention model we will build here is a way of introducing relational reasoning into the model; the goal is not necessarily to distill the data.

The simplest form of absolute attention for an image classifier would be a model that actively crops the image, selecting subregions from the image and only processing those (figure 10.4). The model would have to learn what to focus on, but this would tell us what parts of the image it is using to make its classification. This is difficult to implement because cropping is nondifferentiable. In order to crop a 28 × 28 pixel image, we would need our model to produce integer-valued coordinates that form the rectangular subregion to subset, but integer-valued functions are noncontinuous and thus nondifferentiable, meaning we can't apply gradient descent-based training algorithms.

Figure 10.4. An example of absolute attention where a function might simply look at subregions of an image and only process those one at a time. This can significantly reduce the computational burden, since the dimensionality of each segment is much smaller than the whole image.